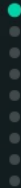




Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy trends across primary energy and electricity · Source: Our World in Data



75% of Global GHG Emissions Still Come from Fossil Fuels

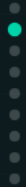
75%

of global greenhouse gas emissions from fossil fuel burning

5M+

premature deaths per year from fossil fuel air pollution

- Since the Industrial Revolution, fossil fuels have dominated the global energy mix.
- Decarbonization requires a rapid shift to low-carbon sources — nuclear and renewable technologies.
- Renewables include solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources.



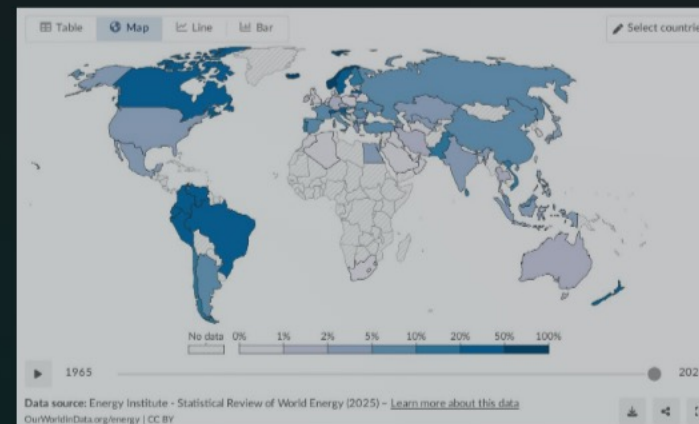
Renewables Now Supply ~14% of Global Primary Energy

2024 data, substitution method · Primary energy = electricity + transport + heating

~14%

of world's primary energy from renewables

- Transport and heating are harder to decarbonize, keeping the overall share lower than electricity alone.
- Norway, Brazil, and New Zealand lead; Middle East and parts of Asia remain below 5%.



Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

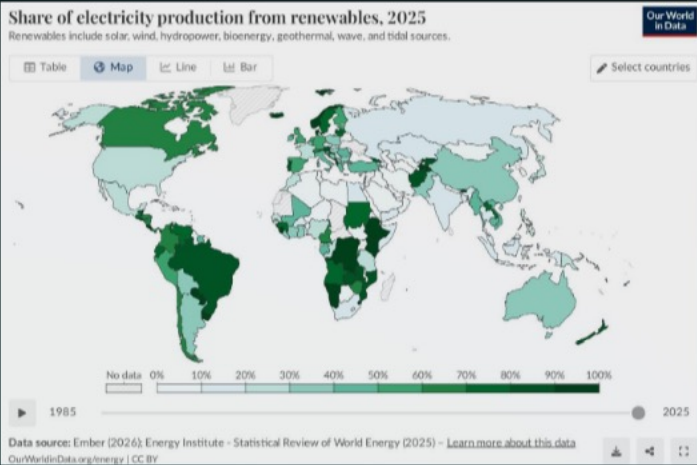


Wind and solar combined now produce nearly as much electricity as hydropower — a dramatic shift since 2010 driven by falling costs and policy support.

TECHNOLOGY	GENERATION	SHARE
Hydropower	~4,500 TWh	~47%
Wind	~2,400 TWh	~25%
Solar	~2,000 TWh	~21%
Other	~700 TWh	~7%

Almost One-Third of Global Electricity Now Comes from Renewables

2025 - Global average ~30%



Share of electricity from renewables by country, 2025

COUNTRY / REGION	RENEWABLE SHARE
Norway	~98%
Brazil	~85%
Canada	~65%
World average	~30%
China	~35%
India	~25%
Saudi Arabia	<2%

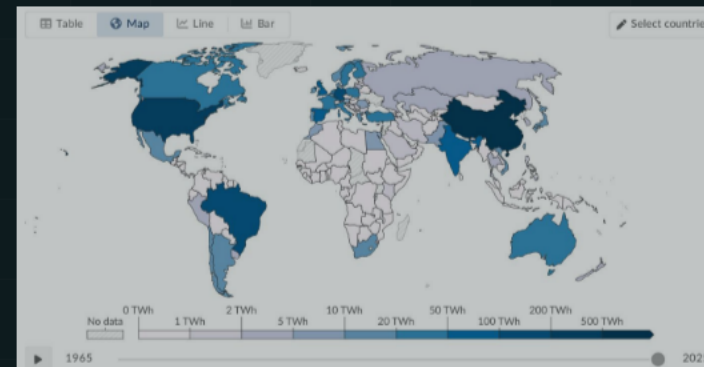
Several African and Middle Eastern countries remain below 10%, reflecting fossil fuel dependence and limited renewable investment.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

~4,500 TWh

global hydropower generation (~47% of all renewable electricity)

- Generating electricity at scale for over a century.
- Top producers: **China**, **Brazil**, and **Canada**.
- Growth potential is limited compared to wind and solar.



Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind Energy

~2,400 TWh

~25% of renewable electricity

Major in Europe, China, and the United States

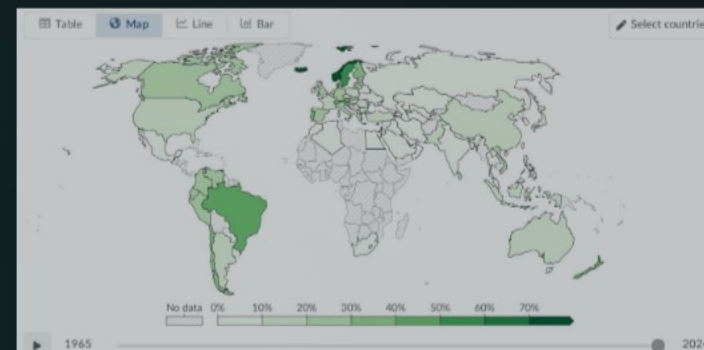
Solar Energy

~2,000 TWh

~21% of renewable electricity

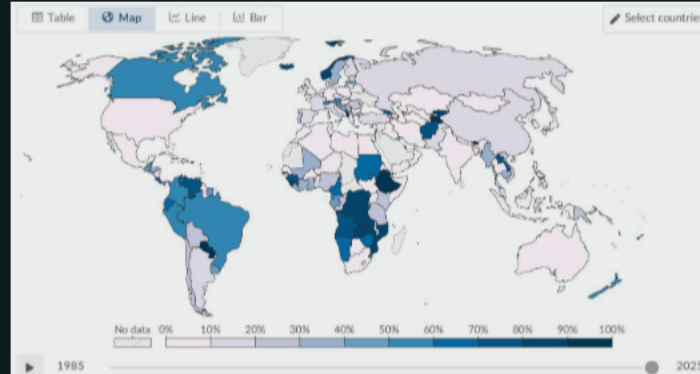
Fastest-growing source since 2010

- Wind and solar **combined now nearly match hydropower** output.
- Solar growth driven by falling costs and policy support.
- Wind includes both onshore and offshore installations.



Wind power generation by country, 2025

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Share of electricity from hydropower by country, 2025

60–98%

Scandinavia, Brazil, Canada — abundant hydropower + wind/solar investment

<10%

Much of Sub-Saharan Africa and the Middle East — minimal renewable infrastructure

- Denmark and Germany have invested heavily in wind and solar.
- Middle Eastern nations remain almost entirely fossil-fuel dependent.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14% primary energy, ~30% electricity — a significant milestone but far from sufficient for climate targets.

Hydropower remains dominant at ~47% of renewable electricity, but its growth potential is limited.

Wind and solar are the fastest-growing technologies in history; combined they now nearly match hydro.

Stark regional disparities persist — Scandinavia and Latin America exceed 60–90%, while much of Africa and the Middle East remain below 10%.

- Transport and heating lag far behind electricity in the renewable transition, keeping overall primary energy share low.

The pace of deployment must accelerate dramatically to meet global climate targets.